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Unique prospects of phase change material Sb_2Se_3 for ultra-compact reconfigurable nanophotonic devices

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Abstract: In this work, we explore inverse designed reconfigurable digital metamaterial structures based on phase change material Sb_2Se_3 for efficient and compact integrated nanophotonics. An exemplary design of a 1×2 optical switch consisting of a 3 µm x 3 µm pixelated domain is demonstrated. We show that: (i) direct optimization of a domain containing only Si and Sb_2Se_3 pixels does not lead to a high extinction ratio between output ports in the amorphous state, which is owed to the small index contrast between Si and Sb_2Se_3 in such a state. As a result, (ii) topology optimization, e.g., the addition of air pixels, is required to provide an initial asymmetry that aids the amorphous state's response. Furthermore, (iii) the combination of low loss and high refractive index change in Sb_2Se_3, which is unique among all phase change materials in the telecommunications 1550 nm band, translates into an excellent projected performance; the optimized device structure exhibits a low insertion loss (~1.5 dB) and high extinction ratio (>18 dB) for both phase states.

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1. Introduction

Silicon-on-insulator (SOI) based platforms have attracted considerable attention for integrated photonic circuits and devices owing to their complementary-metal-oxide-semiconductor (CMOS) compatibility and high refractive index contrast [1–2]. Optical switches are used to control light propagation and play a significant role in optics communication systems. As one of the most fundamental components in optical communication networks, optical switches with a small footprint, low insertion loss (IL), low power consumption, and high extinction ratios are desired in highly integrated photonic circuits and systems [3]. Demonstrated electro-optic (EO) [4–5] and thermo-optic (TO) [6–8] switches require a sustainable external power supply to maintain the switching state, which increases the total power consumption. Moreover, the less than 0.01 refractive index modulation of the switch results in a large footprint, which is detrimental in integrated photonic systems.

Integration of phase change materials (PCMs) with silicon photonic structures overcomes the drawbacks of the aforementioned EO and TO switches. It paves the way for reconfigurable on-chip optical switching operation due to the strong refractive index modulation between amorphous and crystalline states, reversible switching of the states, high switching cycles, and non-volatile property of PCMs [9–12]. As a result, optical switches based on PCMs have been designed and demonstrated for photonic integrated circuit applications [13–17].

$Ge_2Sb_2Te_5$ (GST) has been widely used for phase-change photonics [18–20], and its strong modulation of refractive index between amorphous and crystalline phases enables a significant optical effect. However, the imaginary part of the refractive indices is non-zero in both visible and telecommunication wavelengths and fundamentally limits its further applications due to the resulting absorption. As an alternative to GST, $Ge_2Sb_2Se_4Te_1$ (GSST) [21–22] shows negligible losses in the amorphous phase at 1550 nm, while in the crystalline phase, the imaginary part of

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the refractive index is greater than 0.1, indicating high absorption losses. Overall, the absorption losses of GST and GSST limit their further applications in integrated photonic systems.

In this context, very recently, antimony triselenide (Sb_2Se_3) emerged as a promising PCM due to its extremely low losses ($k < 0.0005$) in both states at a wavelength of 1550 nm [23]. As discussed in [23], the phase of Sb_2Se_3 can be switched by thermal heating or using an optical laser. Through the later effect, non-volatile programmable devices have been recently demonstrated in this materials system, where the PCM phase was locally programmed through optical writing [24]. Based on these recent breakthroughs on Sb_2Se_3 showing it as an effective material for photonic applications, in this work, we explore inverse designed reconfigurable digital metamaterial structures based on this promising PCM for efficient and compact integrated nanophotonics. As an example of the potential of this material, an integrated silicon nanophotonic 1×2 optical switch working in the 1550 nm telecommunications band is proposed. The high refractive index variation between the amorphous and crystalline phases and extremely low losses enables a design with an ultra-compact footprint (3 μm × 3μm), low loss (~1.5dB), and very high extinction ratio (>18dB) between output ports in both phase states. The dense integration of such Sb_2Se_3 ultra-compact nanophotonic devices can pave the way for future non-volatile silicon nanophotonic integrated circuits.

2. Design and simulation

As shown in Fig. 1(a), the 1×2 switch structure consists of an input waveguide, two output waveguides (port 1 and port 2), and a 3 μm × 3 μm switching region on silicon-on-insulator (SOI) platform. Following the recently proposed integrated digital metamaterial concept [25], the switching region is divided into 30×30 square pixels with size 100 nm × 100 nm each. Two

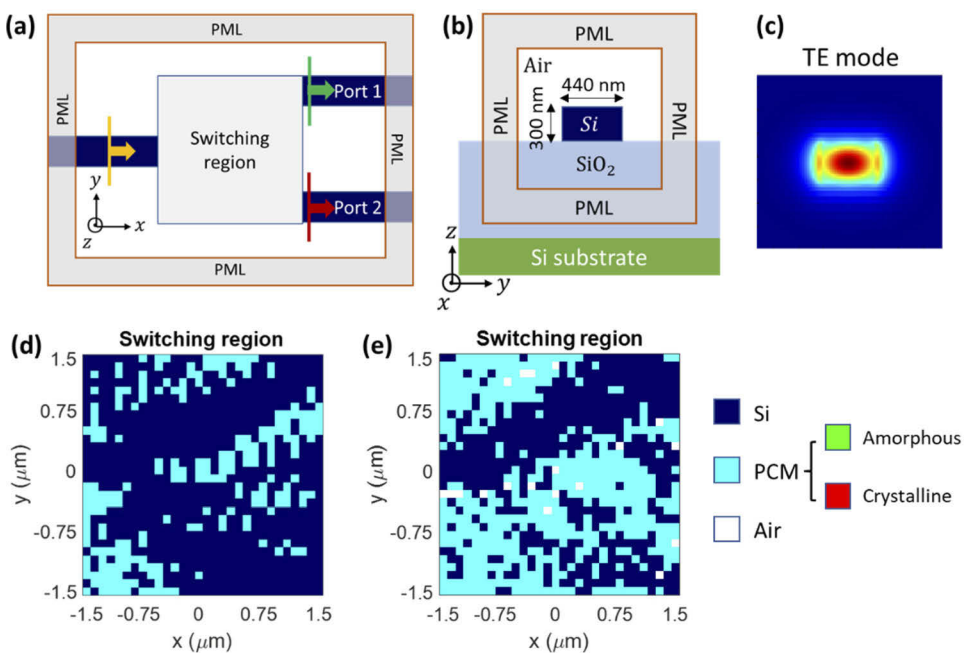

Fig. 1. **(a)** Top view of the proposed 1×2 nanophotonic switch, **(b)** cross-section view of the silicon waveguide, **(c)** the excited TE mode in the input silicon waveguide, **(d)** switching region of a Si-PCM digital metamaterial, which is composed of silicon, and Sb_2Se_3 pixels, **(e)** switching region of a Si-air-PCM digital metamaterial, which is composed of silicon, air, and Sb_2Se_3 pixels.

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families of devices are analyzed: (A) *Si-PCM digital metamaterials*, where each pixel can be either silicon or Sb_2Se_3, and (B) *Si-air-PCM digital metamaterials*, where each pixel can be either silicon, air, or Sb_2Se_3. In both cases, the pixels have the same thickness as the input and output waveguides. The cross-section of the single-mode input and output waveguides is 440 nm (width) × 300 nm (height), as shown in Fig. 1(b). Transverse electrical (TE) polarized light is coupled into the input waveguide, and its mode profile is depicted in Fig. 1(c). The switching region material pixels distributions without and with air pixels, corresponding to (A) and (B), respectively, are shown in Figs. 1(d) and 1(e). At a wavelength of 1550 nm n_{Si}=3.477, and n_{SiO2}=1.444 are used in the design. For Sb_2Se_3, $n_a = 3.285 + 0.000j$, and $n_c = 4.050 + 0.000j$ are employed in the amorphous and crystalline states, respectively [23]. The switch is designed to have one of the output ports dominant mode transmission when Sb_2Se_3 is in the amorphous state, and the other output port dominant mode transmission once Sb_2Se_3 is in the crystalline state. The material distribution of pixels in the switching region is optimized by a modification of the gradient-orient binary search (GBS) algorithm and the direct binary search (DBS) algorithm proposed in [26] and [25], respectively. The GBS and DBS algorithm details are omitted in this discussion due to the elaborated description already provided in previous works. Here, the algorithm was modified so to count the two-phase states of the PCM. Lumerical FDTD [27] was employed to perform the electromagnetic simulations with PML boundary conditions to reduce the reflected wave effect on the simulation accuracy. Details of the algorithm implementation and FDTD simulations are provided in the Supplement 1.

3. Results and discussion

3.1. Si-PCM digital metamaterials

We first analyzed devices consisting of Si and Sb_2Se_3 pixels. Because of the simplicity of the device geometry, which only contains two materials, namely Si and Sb_2Se_3, the devices were optimized employing the GBS algorithm with FOM defined as FOM = | T(port1) – T(port2) | (See Fig. S1 in the Supplement 1 for algorithm flowchart). The evolution (with number of iterations) of the TE mode transmission spectra of the device with Sb_2Se_3 pixels in the amorphous and crystalline state is depicted in Figs. 2(a) and 2(b), respectively. As displayed in Fig. 2, at iteration 218, $FOM_{amo} = 0.3004$, and $FOM_{cry} = 0.8168$. At this point, the maximum value of the sum of FOM_{amo} and FOM_{cry} is achieved. It is noticed that the insertion loss of the dominating output (port 1) is 3.4 dB when the Sb_2Se_3 pixels are in amorphous states. Also, the extinction ratio between transmitted powers at the outputs (port 1 and port 2) is only 4.7 dB. As the Sb_2Se_3 pixels are switched to the crystalline state, the insertion loss of port 2 is 0.85 dB, and the extinction ratio is 22.15 dB. It is noted that low insertion loss and high extinction ratio are desired for both states; however, in the optimized device, the extinction ratio in the amorphous state is low. In this regard, it is important to observe that the refractive index difference between n_a and n_{Si} is Δn_1=0.192, whereas the difference between n_c and n_{Si} is Δn_2=0.573. Therefore, in devices with a small design footprint, as analyzed here, the index contrast in the amorphous state might not be sufficient to lead to substantial asymmetries in transmission. As a result, it is challenging to attain high extinction in the amorphous state. In order to overcome this issue, we propose as a general design principle in cases wherein the index contrast between PCM and Si is limited, to perform an initial step of design topology optimization, e.g., addition of air pixels, so to provide an initial asymmetry and therefore, aid the amorphous state's response. In this case, air pixels are added to the switching region, and their distribution is optimized to provide such initial asymmetry and mitigate the small refractive index contrast between silicon and the amorphous state of Sb_2Se_3.

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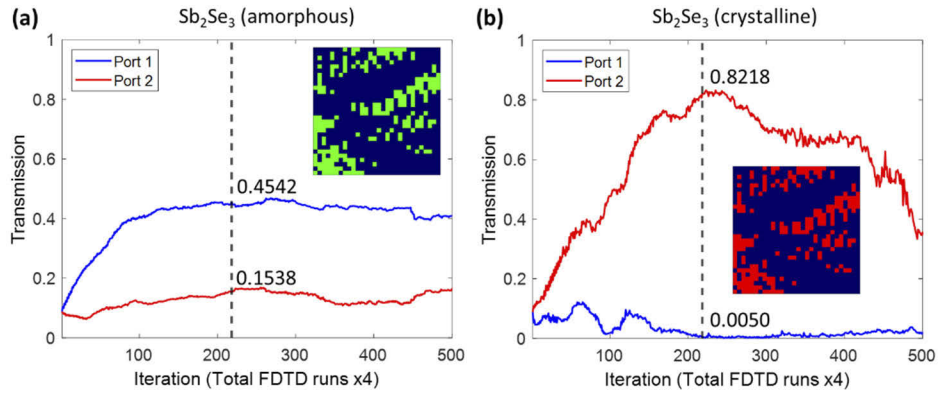

Fig. 2. Optimization of Si-PCM digital metamaterial switch. Evolution of transmission with iterations with Sb_2Se_3 pixels in the **(a)** amorphous state, **(b)** crystalline state.

3.2. Si-air-PCM digital metamaterials

Next, we analyzed Si-air-PCM digital metamaterial structures. Because of the complexity of the structure, now containing three materials, we developed an "objective-guided" three-stage optimization process. An algorithm flowchart of the design process is depicted in Fig. S2 in the Supplement 1. For stage 1, the modified GBS algorithm is performed to optimize the switching region with only silicon and air pixels. As discussed in *Section 3.1*, this is done to provide a starting point where the structure's response is highly asymmetric and guide the overall design of the structure so as to provide a large rejection ratio in the amorphous state wherein the index contrast between Sb_2Se_3 and Si is limited. Therefore, the structure was optimized so that the output port 1 has dominant mode transmission (see Fig. 3(a)). At stage 2, starting from the optimized silicon and air pixels distribution, Sb_2Se_3 pixels are introduced, replacing silicon pixels and leaving the air pixels fixed. GBS is used to simultaneously optimize the output at each port to have high transmission on the corresponding Sb_2Se_3 phase state (see Fig. 3(b)). The optimized pixel distribution is then passed to stage 3 as an initial pattern for DBS optimization to improve the device performance further (see Fig. 3(c)).

The evolution of the transmission through each port versus iterations during each optimization state, together with the optimal results (material pixel distribution and field profile) of each stage, is presented in Fig. 3. For stage 1, the optimized switching region with Si and air pixels (at iteration 20) leads to a TE mode transmission difference of 0.79 as depicted in Fig. 3(a) (left) and also illustrated by the field distribution shown in Fig. 3(a) (right).

The transmission for stage 2 as a function of iterations with Sb_2Se_3 pixels in the amorphous and crystalline state are shown in Fig. 3(b) (left). At the beginning of stage 2 optimization, the distribution of the initial pixels is dominated by silicon and air, which is obtained from stage 1 optimization. It can be observed that the mode transmission difference between port 1 and port 2 is maintained above 0.7 for all the 500 iterations (2000 FDTD simulations) when the Sb_2Se_3 pixels are in the amorphous state, which is attributed again to the limited index contrast between Sb_2Se_3 and Si in such state. The air pixels dominate and mainly set the response of the switching region in this state. As the Sb_2Se_3 pixels are switched to the crystalline state, we observe that the port 1 mode transmission decreases, and the port 2 transmission increases as iteration number increases, which is desired at this stage optimization. At iteration 448, the highest mode transmission difference between port 2 and port 1 is achieved when Sb_2Se_3 pixels are in the crystalline state. In this case, the mode transmission at port 1 is 0.5226, and at port 2 is 0.0521 (difference = 0.4704). With amorphous state at iteration 448, the mode transmission at port 1 is 0.7281, and at port 2 is 0.0065 (difference = 0.7216).

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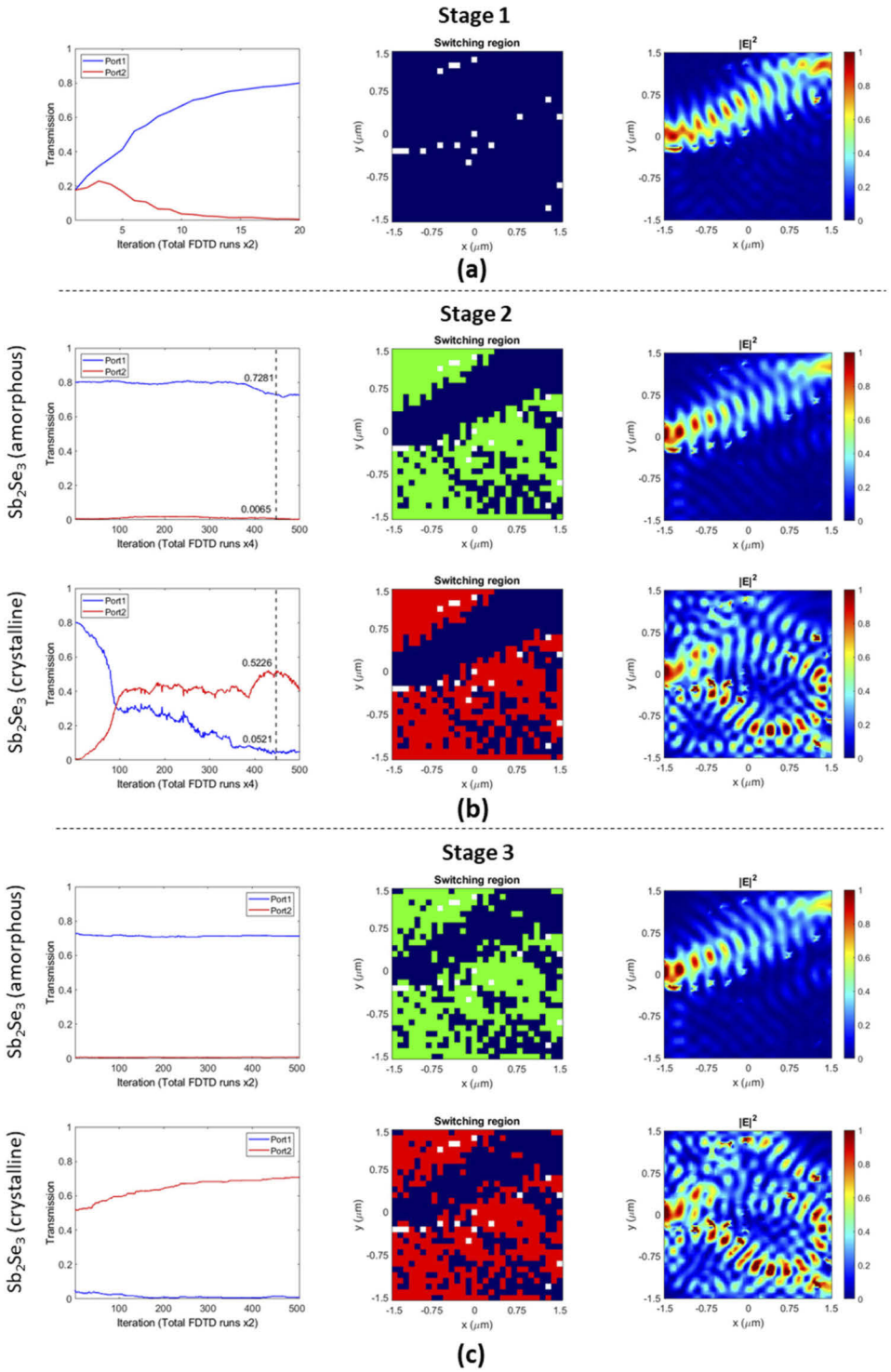

Fig. 3. Optimization of Si-air-PCM digital metamaterial switch. Evolution of transmission with iterations (left), optimized silicon and air pixels distribution (middle), and its corresponding simulated field intensity $|E|^2$ (right), for **(a)** stage 1, **(b)** stage 2, amorphous (top) and crystalline (bottom) states, **and (c)** stage 3, amorphous (top) and crystalline (bottom) states.

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The transmission spectra for stage 3 as a function of iterations with Sb_2Se_3 pixels in the amorphous and crystalline state are shown in Fig. 3(c) (left). At stage 3, the transmission differences between the two output ports in both states are optimized to be greater than 0.7 (termination condition). In the amorphous state, upon convergence, the mode transmission at port 1 is 0.7118, and at port 2 is 0.0076 (difference = 0.7042). In crystalline state, the mode transmission at port 1 is 0.0094, and at port 2 is 0.7107 (difference = 0.7013). The optimization, in this case, ended due to the satisfaction of the termination condition. From the electric field intensity distribution of the final optimized device, depicted in Fig. 3(c) (right), it is clearly observed that the mode light is transmitted to either port 1 or port 2 when the Sb_2Se_3 pixels are either on amorphous or crystalline states, respectively. It is worth noticing that devices with superior performance are possible if the termination condition is modified to, e.g., 0.8 (or even higher), but it will take a much longer time (number of iterations) to terminate.

The final optimized device was simulated in a wavelength range from 1500 nm to 1600 nm for both amorphous and crystalline states. The dispersion data of Sb_2Se_3 in this wavelength range, shown in Fig. 4(a), is obtained from [23] and imported to the Lumerical material database for 3D FDTD simulation of the final optimized pixel distribution. As shown in Fig. 4(b), with Sb_2Se_3 pixels in the amorphous state, the optimized device shows a broadband wavelength operation. The extinction ratio at the center wavelength (1550 nm) is 19.7 dB. For the crystalline state, depicted in Fig. 4(c), the extinction ratio between output ports is 18.6 dB. However, it is observed that the device exhibits a narrowband operation in this state. The insertion loss in both states is ~1.5dB. It is to note that this performance in terms of extinction ratio and insertion loss is similar to that reported in 1×2 switches based on GST (e.g., ~4dB IL, 7dB rejection [28]; ~1dB IL, >10 dB rejection [29]; 2.4dB IL, 7dB rejection [15]); however, the footprint of the devices here proposed is >5× smaller than that of these devices (length = 3 μm herein versus 15 μm in [28], 30 μm in [29], and a ring-resonator diameter of 30 μm in [15]). Furthermore, in this work all three stages of optimization were performed at a single wavelength of 1550 nm. In principle, the operational bandwidth for the crystalline state could be improved if optimization was to be performed in a wavelength band rather than at a single wavelength.

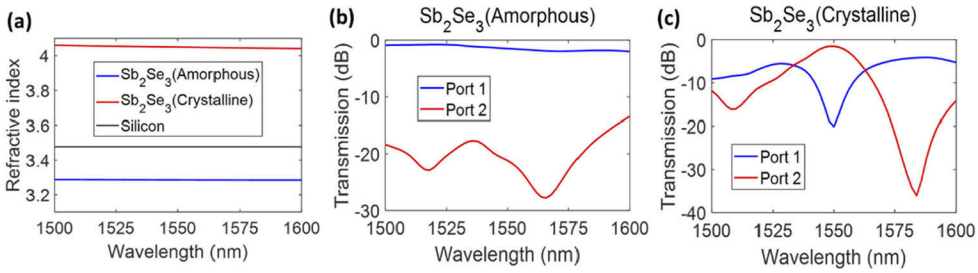

Fig. 4. **(a)** Refractive indices of silicon and Sb_2Se_3. Transmission spectra of the optimized device with Sb_2Se_3 pixels in the **(b)** amorphous, and **(c)** crystalline state.

We would like to point out that the dimensions of the individual pixels that have been employed here in the patterns can be fabricated using electron-beam lithography (EBL), focused-ion-beam lithography (FIB), or projection photolithography. We have demonstrated this in the past [25,30]. Fabrication of the overall device could start by the definition of the switching region and input/output waveguides, followed by an EBL step to define and etch the regions consisting of PCM pixels, followed by Sb_2Se_3 deposition (either through evaporation, sputtering, or atomic layer deposition) and lift off. Followed finally by capping to prevent oxidation and loss of Se, and a final step of lithography, e.g., FIB, to define the air pixels. It is also possible to carry out a different approach, in which is first the PCM deposited, followed by EBL to define the PCM pixels, then deposition of silicon to define the switching region, polishing, and finally FIB to

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define the air pixels. A similar process has been recently reported in [31] for the realization of meta-displays based on Sb_2Se_3 where unit cells with PCM domains with minimum features on the order of 110 nm have been demonstrated. It is to note that the addition of a capping layer will, in general, affect the operation of the device; however, the presence of such layer can be in principle considered when optimizing the device geometry. Furthermore, if the capping layer is thin, e.g., 10nm or below, and consists of a low refractive index material, we do not expect results to deviate significantly from those of the device herein optimized. Another important metric for the proposed devices will be in relation to its switching speed. In this regard, it is worth mentioning that although early works had shown relatively slow amorphous to crystalline switching times (>100ms [23]), crystallization of Sb_2Se_3 with 100µs pulses has been recently experimentally demonstrated [32].

4. Conclusions

A compact (3 µm× 3 µm) integrated silicon nanophotonic 1×2 optical switch was proposed and optimized based on the modified gradient-orient binary search and direct binary search algorithms. The designed switch device consists of Sb_2Se_3 phase change material, which tailors the device response when the material is switched between its amorphous and crystalline states. The optimized optical switch exhibits a low insertion loss of ~1.5 dB and high extinction ratio >18 dB for both amorphous and crystalline phases. This work shows the potential of inverse designed reconfigurable digital metamaterial structures based on phase change material Sb_2Se_3 for efficient and compact integrated nanophotonics and presents general design strategies for such devices.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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